Fixing the long tail

Process improvement for NYC street-maintenance requests (311, 2025)

212,103 DOT service requests · Python + DuckDB SQL · root-cause analysis · to-be process · impact model

Mohammad Amir Khan · Independent case study on public data · October 2026

Source: NYC Open Data, 311 Service Requests from 2020 to Present (erm2-nwe9). Not affiliated with or endorsed by NYC DOT or NYC 311.

The problem

Residents wait a median of 1.9 days, but 1 in 10 waits 18+ days, and much of DOT's field effort finds nothing to fix

212,103

DOT service requests created in 2025

2.1M

request-days of waiting (sum of time to close)

39%

of Street Condition requests are failure demand

46%

of requests close without a published outcome

Business questions

- Where does residents' waiting time accumulate, and why?
- How much inspection effort is spent on requests DOT cannot or need not fix?
- Which process changes would cut waiting and wasted visits, and by how much?

Scope

- In: all 311 requests assigned to NYC DOT, created 1 Jan - 31 Dec 2025
- Out: field crew rosters and costs (not public); requests handled outside 311
- Lens: Lean, i.e. lead time, failure demand, first-time-right

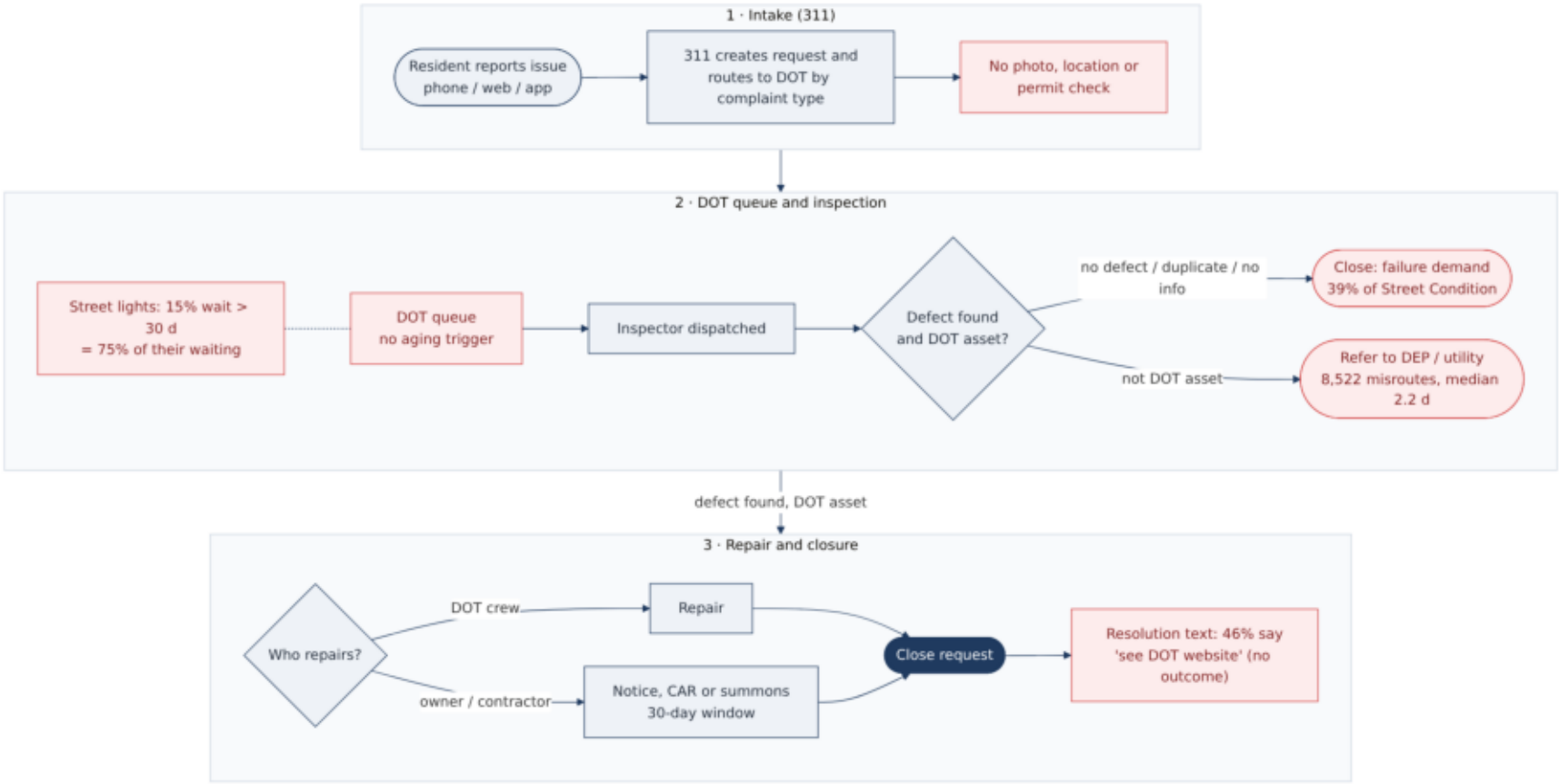
Data and method

A reproducible pipeline: download → DuckDB SQL models → tests → charts → this deck

- Source: NYC Open Data '311 Service Requests from 2020 to Present' (erm2-nwe9), via the public SODA API. Published without license or use restrictions (NYC Admin. Code §23-502(d)).
- Snapshot: 2026-10-17 (New York time); 212,103 rows, SHA-256 in data/raw/manifest.json.
- SQL (DuckDB): staging → request facts → outcome classification (9 classes from DOT's free-text resolution) → 10 analysis queries exported to CSV.
- Request-days = sum of (closed − created). It weights each request by how long a resident waited.
- Failure demand = closed as no defect found, wrong agency, duplicate or insufficient information.
- Data quality: 835 records closed before they were created (excluded); 5,047 still open at snapshot (reported separately as backlog).

As-is process

Pain points (red) come from the data: misroutes, no-defect visits, no aging trigger, unpublished outcomes

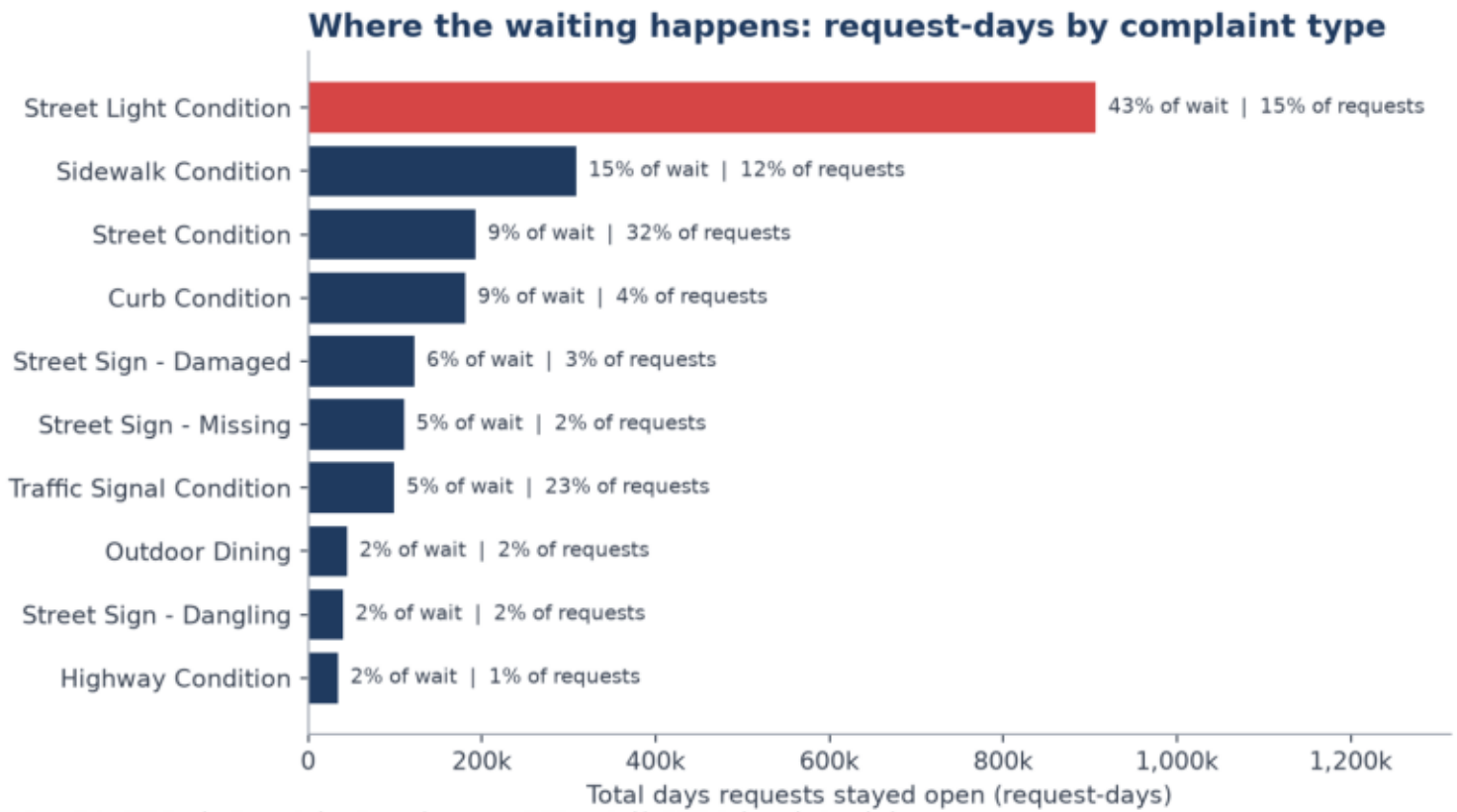


- Intake accepts any report as-is: no photo, asset-owner or permit check.
- The inspector visit is the first filter, so no-defect and wrong-agency jobs still cost a visit.
- No aging trigger: old street-light outages sit in the queue.
- Closure text often hides the outcome, so first-time-fix cannot be measured.

Where the waiting happens

Street lights: 15% of requests, 43% of all waiting time

Street Light Condition is 15% of requests but 43% of all waiting time. Top 4 types = 75% of request-days.



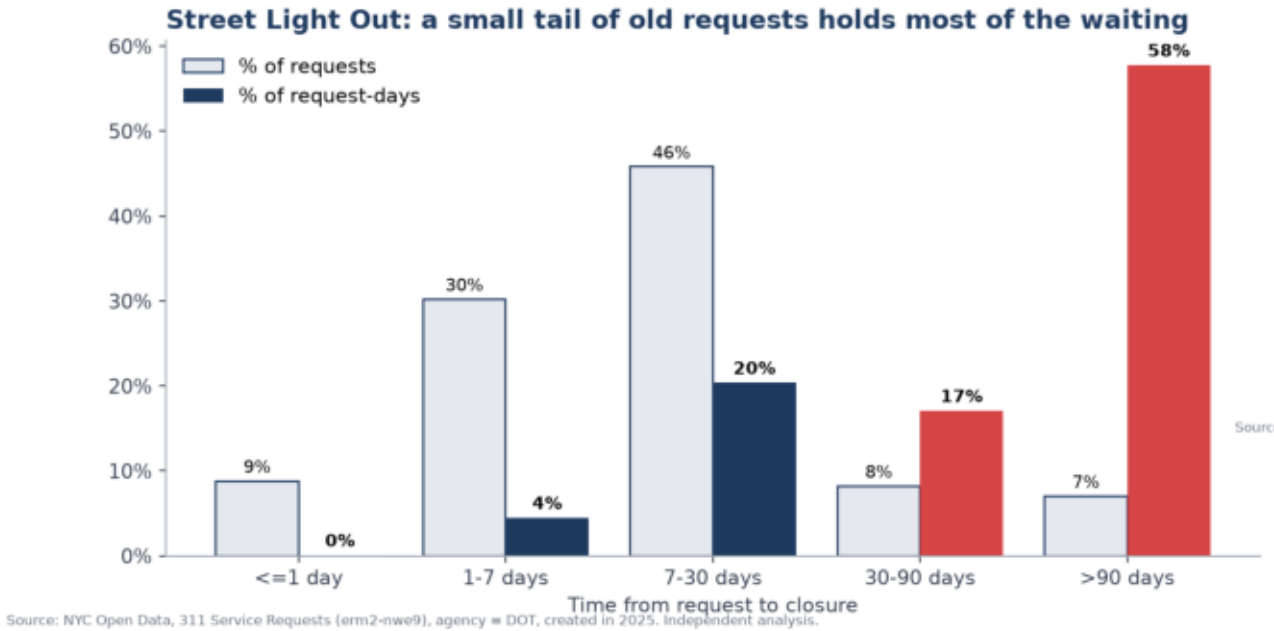
Source: NYC Open Data, 311 Service Requests (erm2-nwe9), agency = DOT, created in 2025. Independent analysis.

- The top 4 complaint types hold 75% of request-days.
- Potholes are a success story: 71% repaired, median 0.9 days.
- So the biggest lever for waiting time is street-light outages, not pothole speed.

Street Light Out: the long tail

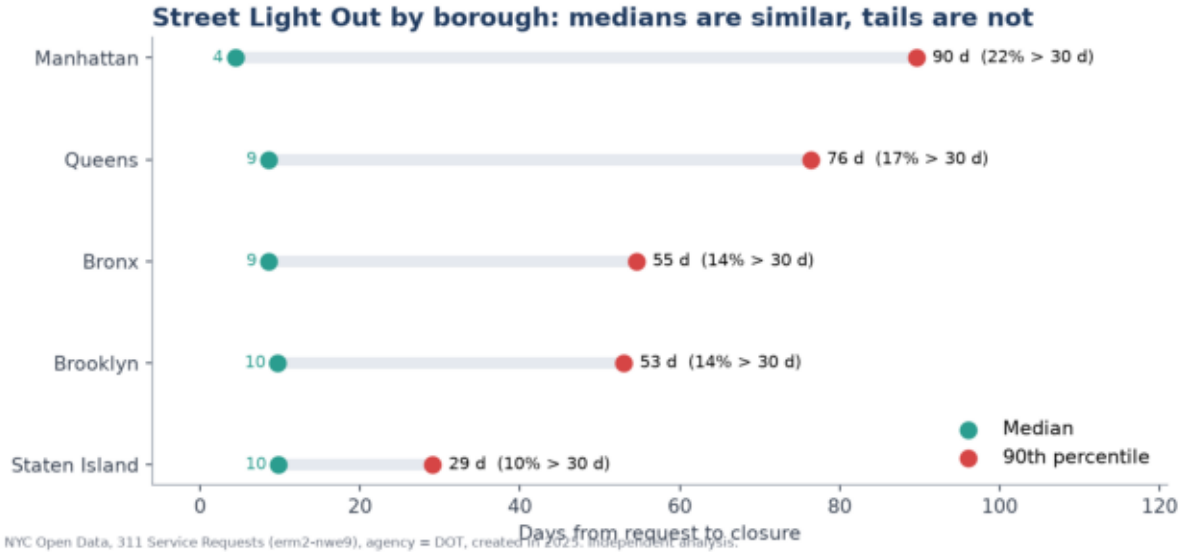
15% of outages wait more than 30 days and hold 75% of street-light waiting

Requests open more than 30 days are 15% of the volume but 75% of request-days (n = 24,380 closed requests).



Source: NYC Open Data, 311 Service Requests (erm2-nwe9), agency = DOT, created in 2025. Independent analysis.

Manhattan has the fastest typical fix (4 d) but the longest tail (p90 90 d) vs Staten Island (p90 29 d).



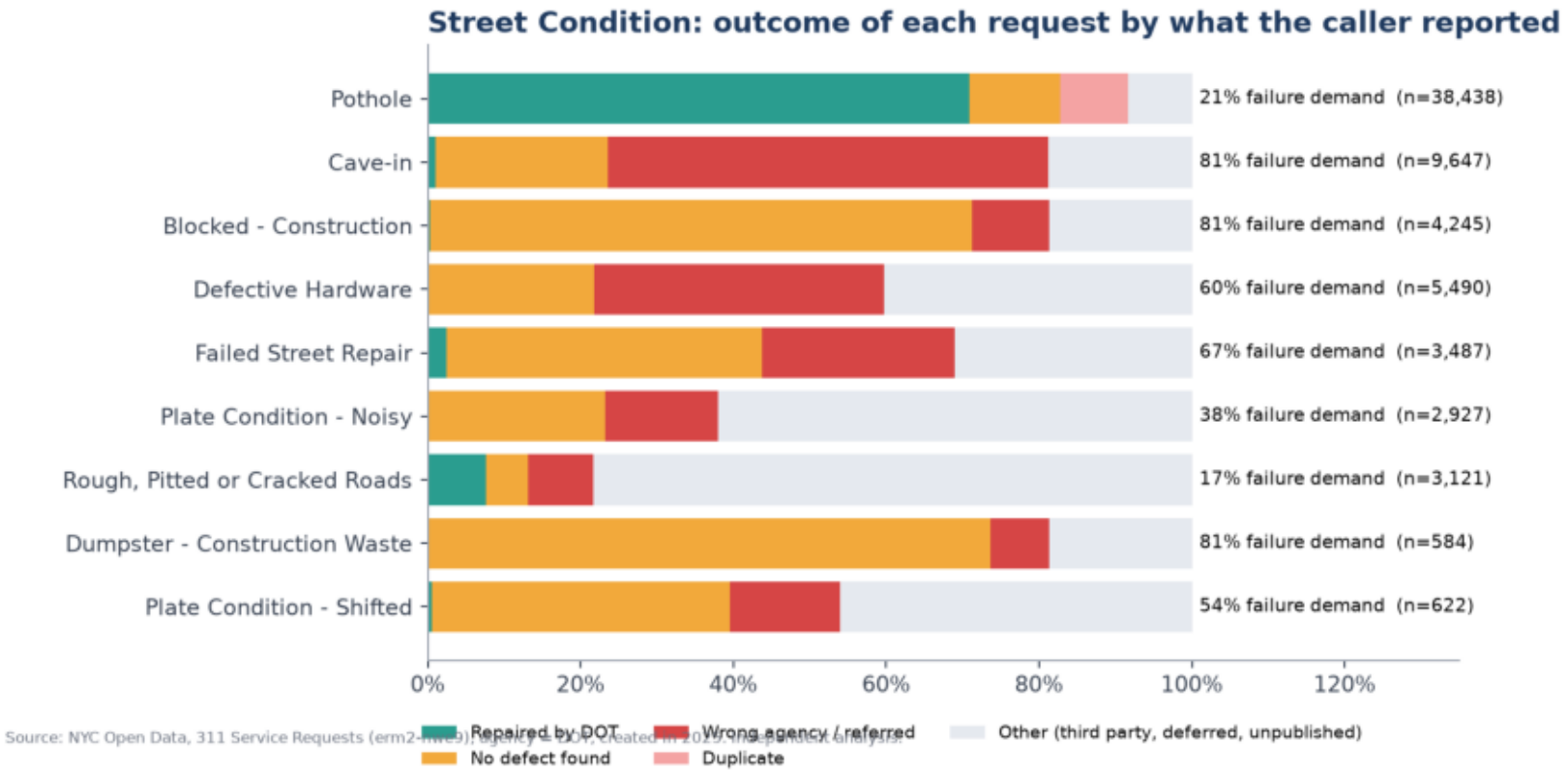
Source: NYC Open Data, 311 Service Requests (erm2-nwe9), agency = DOT, created in 2025. Independent analysis.

- Manhattan: fastest median (4 d) but p90 90 d; Staten Island p90 29 d.
- Jan-Apr requests: p90 62-86 d vs 40 d in July.

Failure demand in Street Condition

27,506 of 70,330 requests (39%) end without a DOT repair being possible

Potholes are fixed 71% of the time. Cave-ins, hardware and construction complaints mostly end with 'no defect' or a referral to another agency.



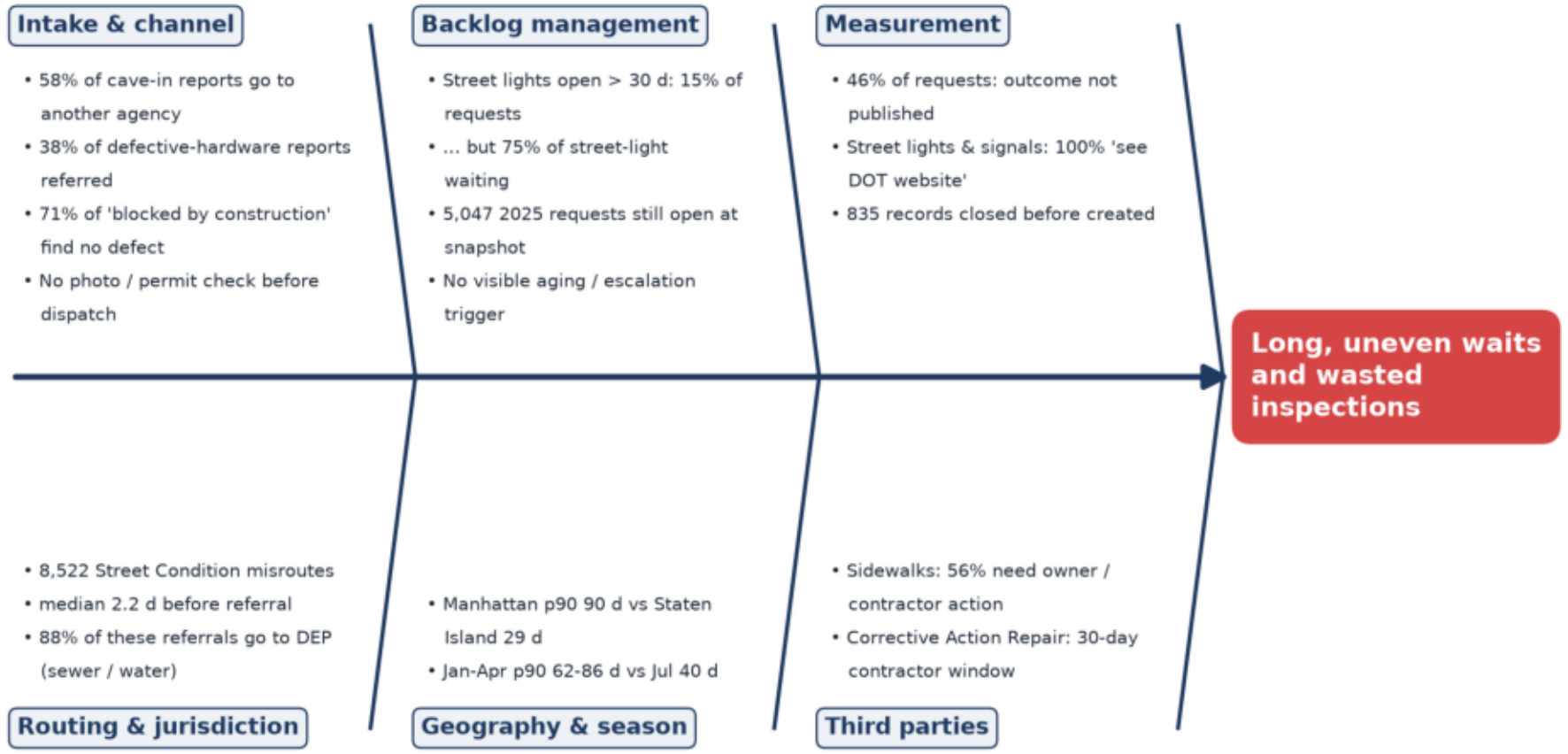
Source: NYC Open Data, 311 Service Requests (erm2-nwe9), agency = DOT, created in 2025. Independent analysis.

- Cave-in: 58% referred elsewhere.
- Defective hardware: 38% referred.
- 88% of these referrals go to DEP (sewer and water).
- Blocked - Construction: 71% 'no defect', often a permitted work site.

Root causes

Two process gaps explain most of it: intake has no screening, and nobody manages the aging backlog

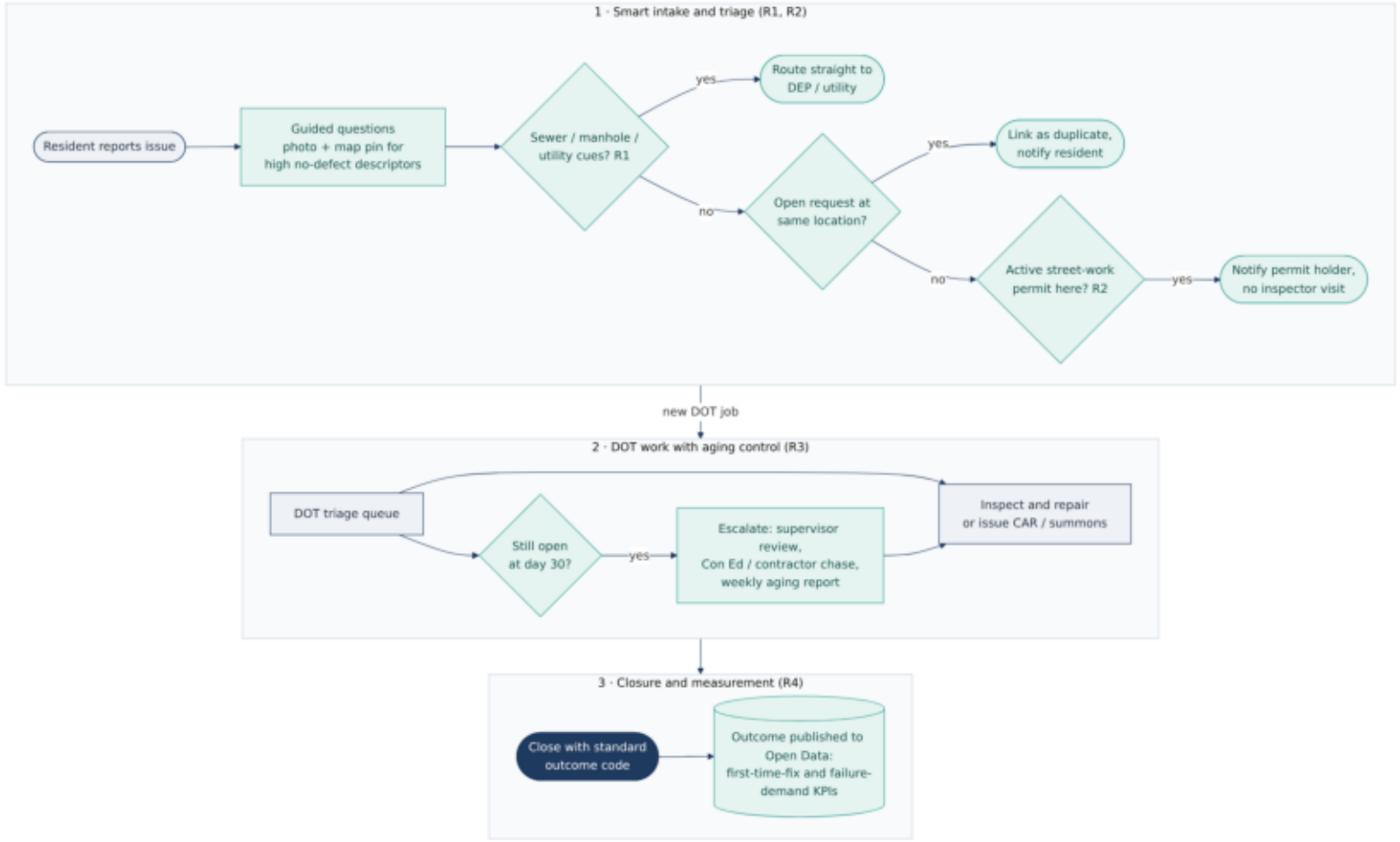
Root causes (Ishikawa), each backed by a number from the 2025 data



Source: NYC Open Data, 311 Service Requests (erm2-nwe9), agency = DOT, created in 2025. Independent analysis.

To-be process

Screen and route at intake, escalate at day 30, publish an outcome on every closure



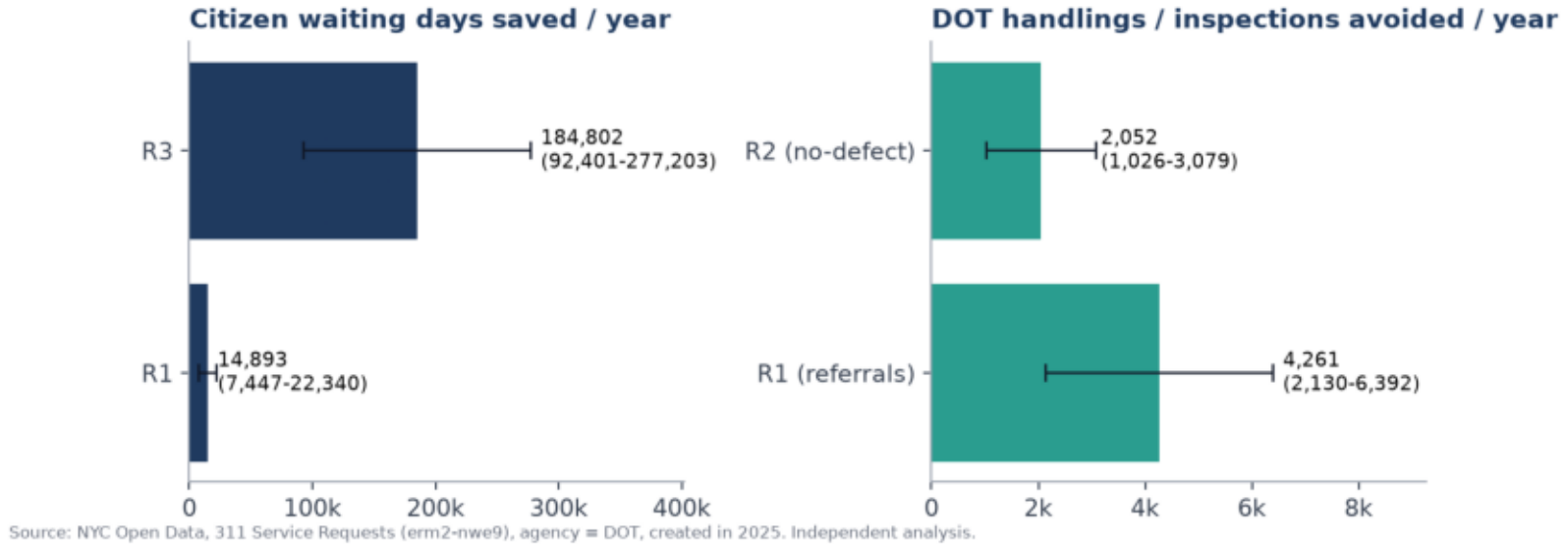
- R1: guided questions send sewer, manhole and utility conditions straight to DEP or the utility.
- R2: photo + map pin, a duplicate check and an active-permit check run before an inspector is sent.
- R3: any street-light outage still open on day 30 is escalated and listed in a weekly aging report.
- R4: every closure carries a standard outcome code that is published to Open Data.

Recommendations and estimated impact

Base case: 199,695 citizen-days and 6,313 DOT handlings avoided per year

	Change	Addressable (2025)	Base-case impact / year
R1	Route sewer / utility cues at intake	8,522 misroutes	**4,261 handlings, 14,893 citizen-days**
R2	Photo + permit check before dispatch	5,131 no-defect visits	**2,052 visits (~3,079 h*)**
R3	Day-30 escalation for street lights	3,717 outages > 30 d	**184,802 citizen-days; p90 61→45 d**
R4	Publish outcome code on every closure	97,937 unpublished	**Outcome measurable for 100% of requests**

* inspector-hours use an illustrative 1.5 h per visit (not in the data). Low/high ranges and all assumptions are in assumptions.toml.



Roadmap, KPIs and caveats

Start with the cheap, measurable changes; prove impact before scaling

90-day roadmap

- Days 0-30: publish outcome codes (R4); weekly street-light aging report (R3).
- Days 30-60: pilot routing questions for cave-in / hardware in one borough (R1).
- Days 60-90: photo + permit check for construction-related descriptors (R2); compare pilot vs control boroughs.

KPIs to track

- Street-light p90 days and % open > 30 days
- Failure-demand rate by descriptor
- Misroute rate and days to referral
- % of closures with a published outcome code

Caveats

- Public 311 data only: no crew capacity, cost or asset data, so impacts are in days and visits, not dollars.
- Closure time can reflect paperwork as well as repair. Outcome classes come from DOT's free text using documented rules.
- Estimates assume 2025 volumes repeat. Pilot results should replace the assumed adoption rates.